GPS Map Matching

An HMM-Based Approach for Mapping Noisy GPS Trajectories to Real Road Networks



GeoLife
GPS Trajectories

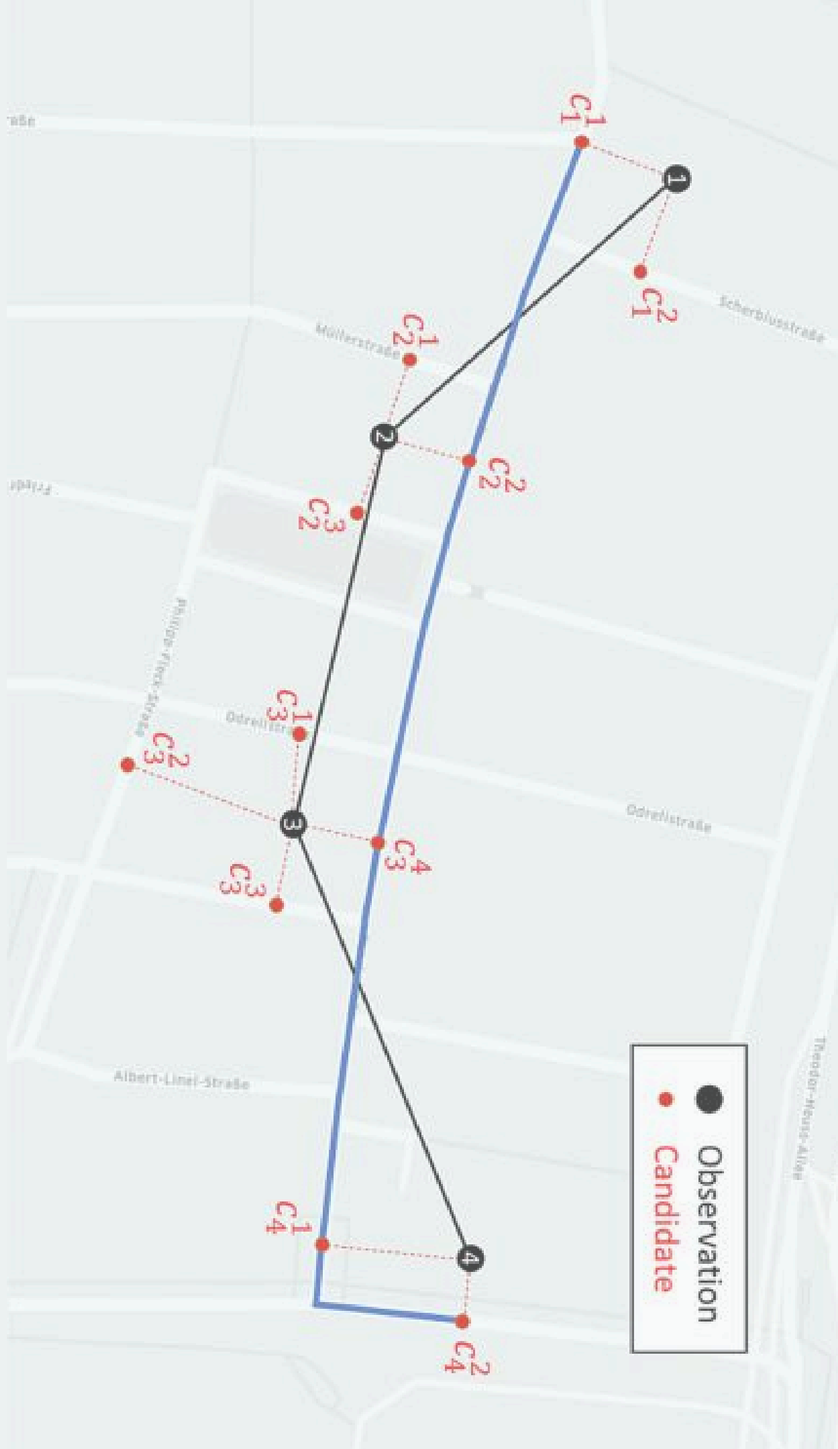
OpenStreetMap
Road Network

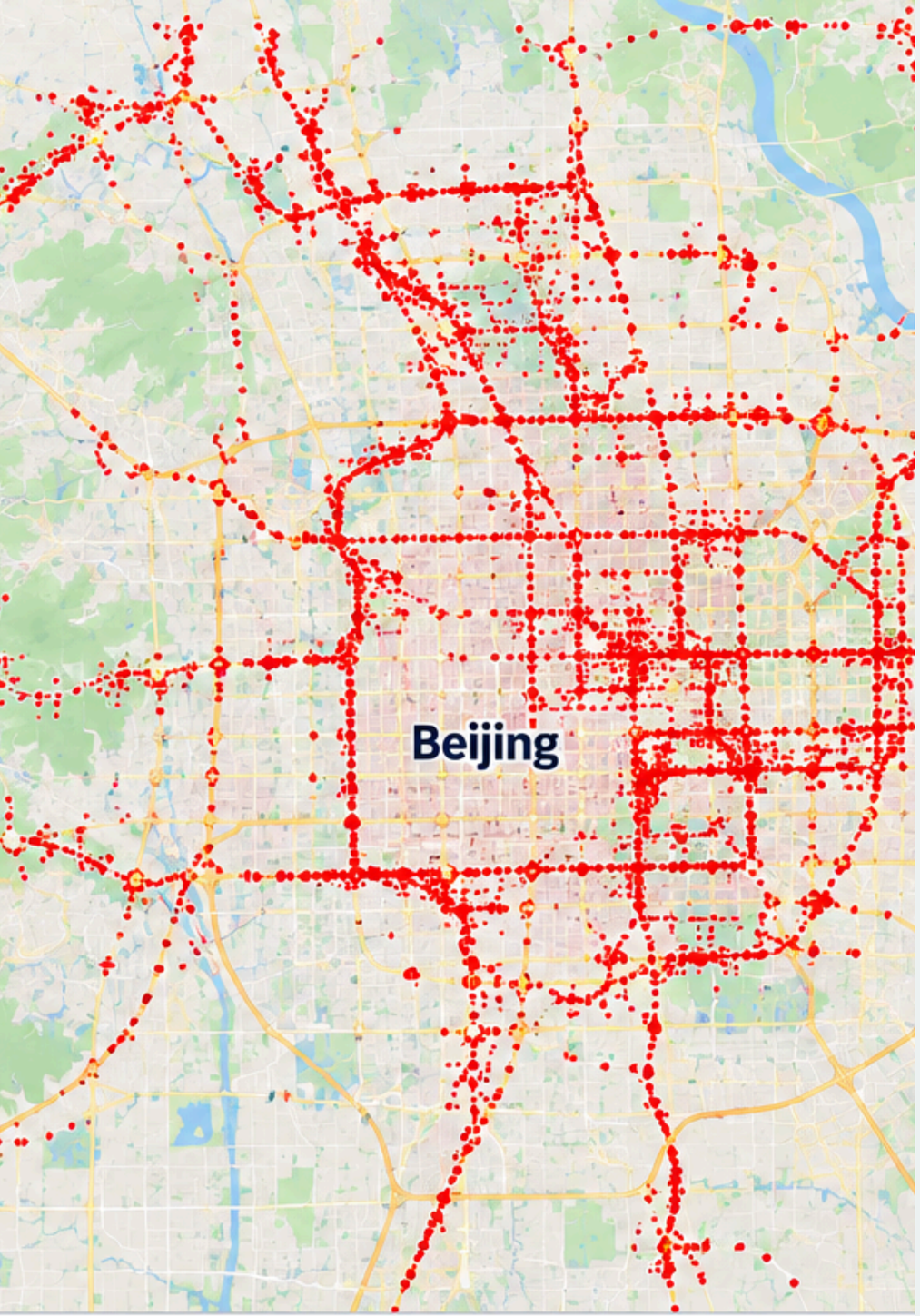
HMM + Viterbi
Accurate Road Matching

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Problem Statement

- GPS devices often **produce noisy or inaccurate location** coordinates that **do not align precisely with actual roads**. This makes it difficult to determine the **roads traversed using raw GPS trajectories alone**. The problem is to identify the most likely road segments corresponding to GPS observations by accounting for positional errors, road-network connectivity, and movement consistency.



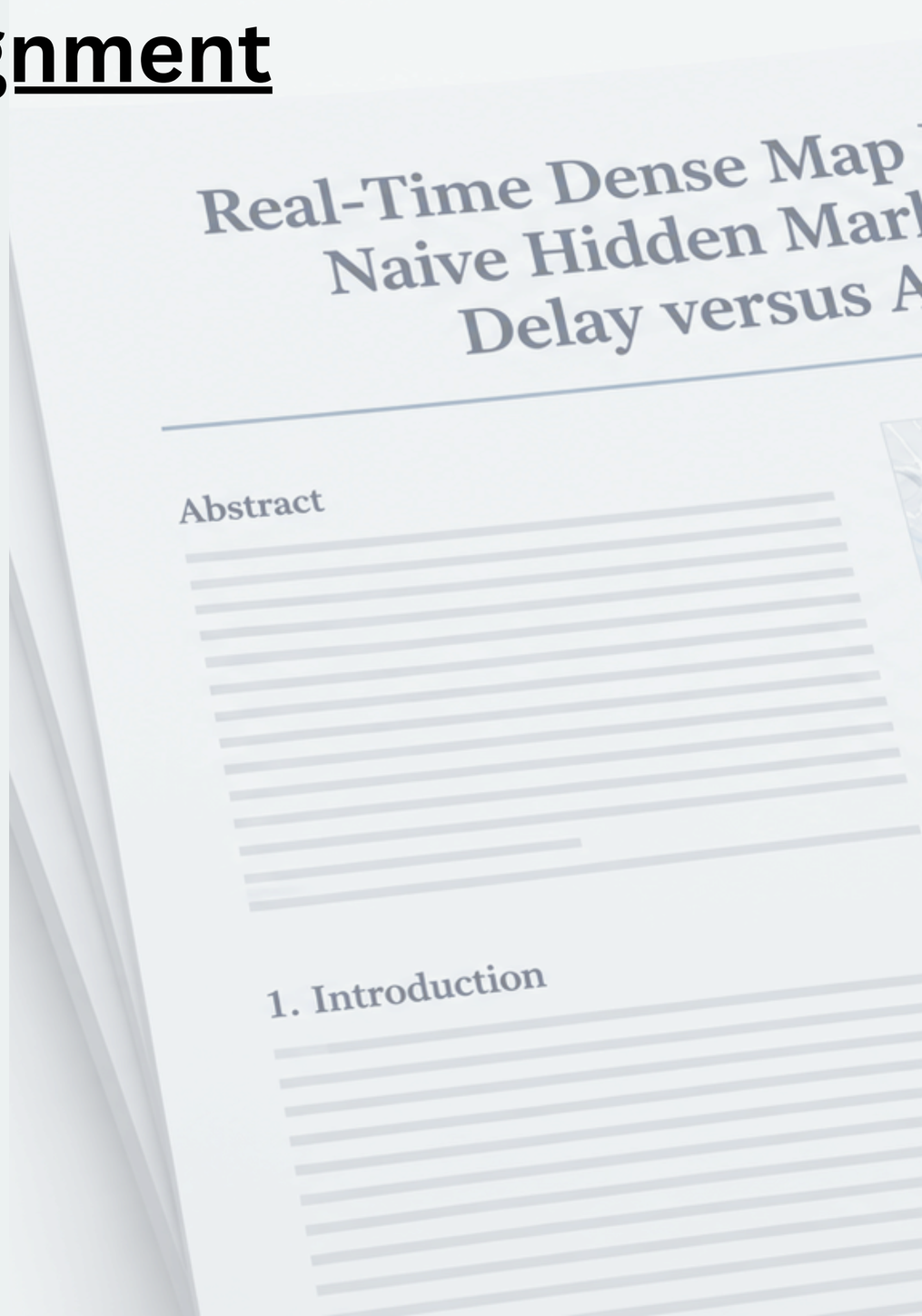


Dataset Details

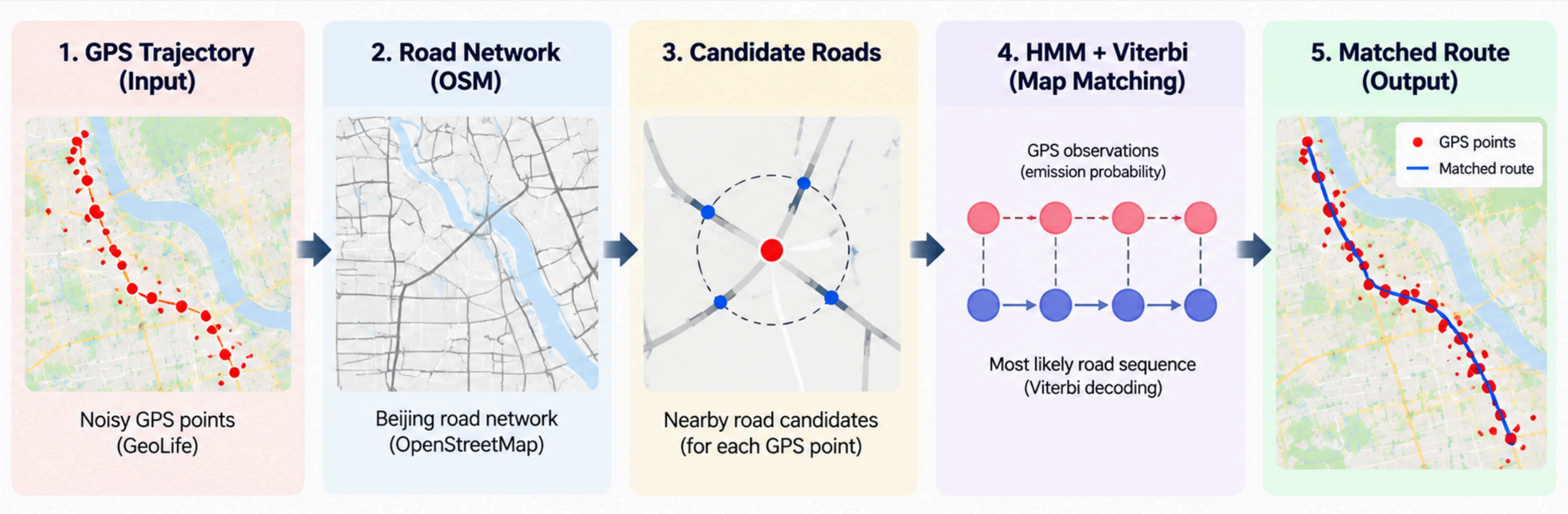
Dataset: GeoLife GPS Trajectories 1.3
Source: Microsoft Research Asia
Data type: Real-world GPS trajectory data
Geographic coverage: Primarily Beijing, China
Users: 182
Trajectories: Approximately 18,670
GPS observations: Approximately 24.8 million
Data attributes: Latitude, longitude, altitude, date and time
Preprocessing: GPS records were cleaned, timestamped, and organized into trajectories and segments.
Road network: OpenStreetMap (OSM) driving-road network for Beijing
Application: Candidate road generation, HMM probability scoring, and Viterbi-based map matching.

Approach & Paper Alignment

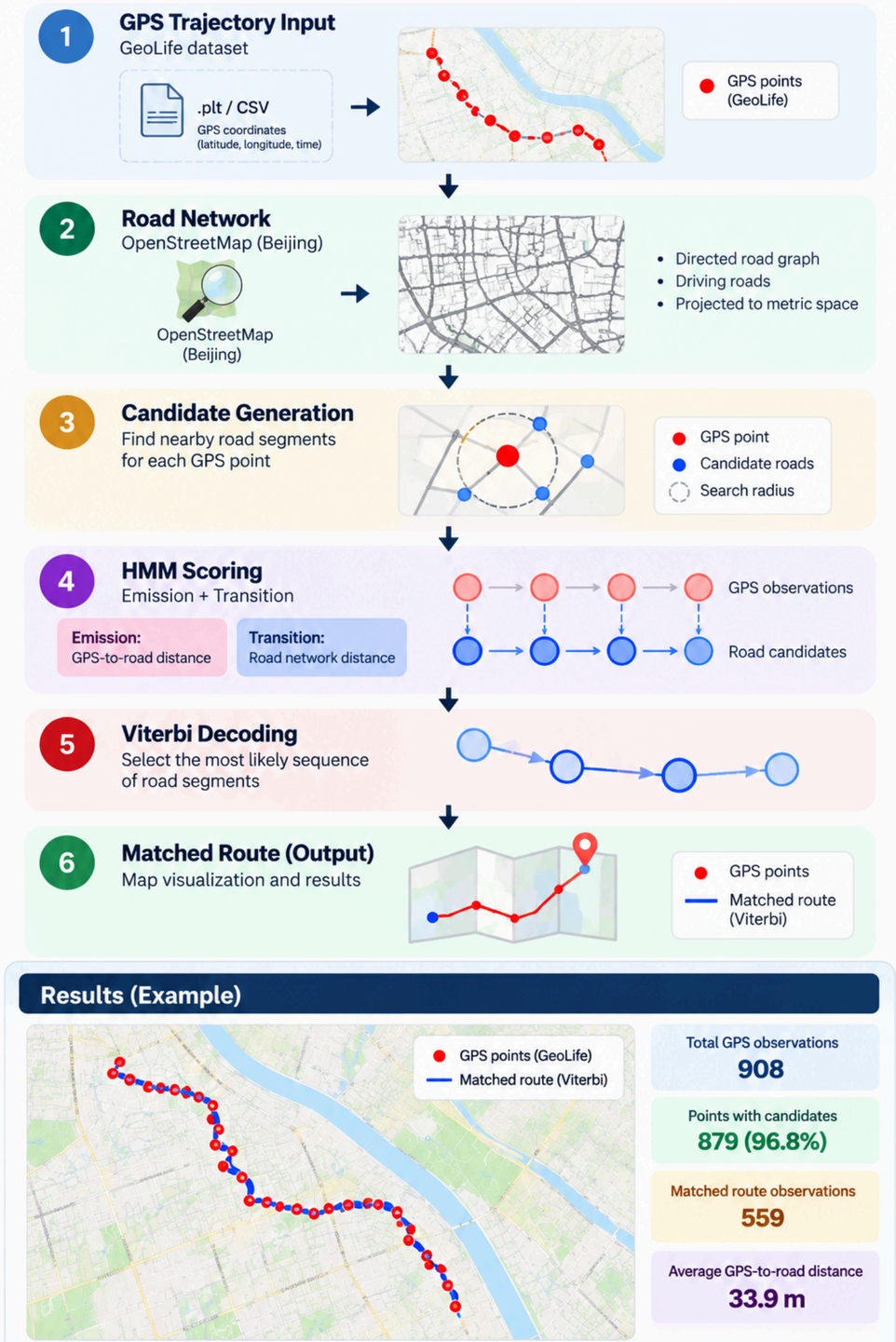
- **GPS Input:** GeoLife trajectory dataset.
- **Road Network:** OpenStreetMap (OSM).
- **Candidate Generation:** Identify nearby road segments for each GPS point.
- **HMM + Viterbi:** Use emission and transition probabilities to select the most likely road sequence.
- **Paper Alignment:** Follows the paper's core HMM-based map-matching principle.
- **Key Difference:** Our implementation focuses on offline trajectory matching; the paper explores real-time matching and the delay–accuracy trade-off.



Implementation



The system **processes raw GPS trajectories** from the GeoLife dataset and **identifies nearby road segments using the OpenStreetMap road network**. An **HMM** evaluates **candidate roads using emission and transition probabilities**, while the **Viterbi algorithm selects the most likely sequence of road segments**. The final matched route is visualized by comparing the original GPS points with the predicted road path.



Results

- Processed GPS trajectory containing **908** observations from the GeoLife dataset.
- Generated road candidates for **24,876,978** points wherever feasible, using the **Beijing OpenStreetMap** road network.
- Applied **HMM emission** and transition scoring to evaluate GPS-to-road proximity and movement consistency.
- Used the **Viterbi algorithm** to identify the **most likely sequence of road segments**.
- Produced a matched route containing **559 observations** and a map visualization comparing the original GPS trajectory with the predicted route.

Thank You